

how do ai data centers use up water?

Posted by u/Consistent-Garlic-78 • 0 points • 8 comments

okay so i understand that they use water for data center cooling, and i assume its fresh drinkable water from how people discuss it right. but i dont understand how it "uses it." does it not just evaporate and go back into the water cycle, so it can later be used for other things? does it become undrinkable?

Comments

u/Lumpy-Notice8945 • 3 points • 2025-09-18 13:32 UTC

This is not specific to AI, google search data centers or amazons data centers can be cooled with water too, they evaporate water like powerplants do to cool the computers inside.

u/herne_hunted • 1 points • 2025-09-18 14:34 UTC

A bit unfair to power plants. Most are built by rivers or the sea so that they're not using potable water for cooling.

u/Lumpy-Notice8945 • 1 points • 2025-09-18 14:36 UTC

How is that different to servers/data centers? Most of these are not cooled by water at all and the ones that are are mostly near natural water sources.

u/herne_hunted • 2 points • 2025-09-18 14:45 UTC

I'm no an expert on data centres but I know there are several in the middle of Birmingham, a city which doesn't have a river running through it. Are they running on boreholes?

u/Lumpy-Notice8945 • 1 points • 2025-09-18 14:56 UTC

There is data centers in every bigger city. But they are not cooled with water for the most cases, they just use regular ACs. That drains electricity and thats it. To actually need water cooling you need to be in the top 10% of size in data centers, most dont need that and its way too much work compared to ACs or literally just blowing cool air in the room.

u/herne_hunted • 1 points • 2025-09-19 06:46 UTC

Thatnks for the explanation.
Obvious in hindsight, if you're going to build a really big data

centre then you're not going to find enough space in a city centre anyway. Must need some impressive air con systems. I'll have to look more closely next time I'm passing.

u/Thanks_Its_new • 0 points • 2025-09-18 13:15 UTC

The water cycle takes a certain amount of time to complete and it is difficult to change the timeline of that on a large scale. Additionally sources of ground water like in aquifers can take 100s and 1000s of years to replenish. So the way it's being done now is that we are using water at a faster rate than is being naturally replenished by the water cycle. There is probably a way to close this loop for cooling but that costs money which is antithetical to the purpose of AI which is to only make money at the expense of others.

u/rhomboidus • 2 points • 2025-09-18 13:11 UTC

They turn potable water into non-potable water.

Nobody is concerned that water itself is being "used up" and disappearing or whatever. People are concerned that the Plagiarism Machine is using up more potable water than a small city, which raises the price of water for actual humans.